23CSE202 DATABASE MANAGEMENT SYSTEMS
S3 B.Tech CSE
Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

3. Database Design Document

Group No :C2
Team Members :

Roll No	Student Name
AM.SC.U4CSE25205	Arjun S P
AM.SC.U4CSE25217	Gayathri Rajeev
AM.SC.U4CSE25235	N Samiksha
AM.SC.U4CSE25247	Sarvesh Sudheer

Project Title : Agriculture Supporter

3.1. Detailed description of the project

Agriculture Supporter is a database-backed web application designed to give farmers a single platform for accessing structured crop-production and animal-husbandry information. The current UI is organized around registration/login, a post-login dashboard, Animal Husbandry, Crop Production, Search by Location, and an authenticated AI chatbot.

The system begins with user registration and login. Authentication is database-based rather than using hardcoded credentials. A User record stores the information needed to create an account and authenticate the user. After successful login, the user reaches the dashboard containing three main information modules: Animal Husbandry, Crop Production and Search by Location. The UI also provides a floating AI chatbot on authenticated pages.

The Animal Husbandry module is hierarchical. The first page displays farm/animal categories such as Poultry, Dairy, Goat Farming, Sheep Farming, Piggery and Aquaculture as image-based navigation cards. Selecting a category displays the animals under that category. Selecting an individual animal opens a detailed page containing care, feeding, diseases and prevention, housing, breeding and management information. Therefore the database represents both livestock categories and individual livestock records.

The Crop Production module follows the same hierarchy. The UI displays categories such as Cereals, Pulses, Vegetables, Fruits, Plantation Crops, Spices and Organic Farming. Selecting a category displays the crops belonging to it, and selecting a crop opens a detailed cultivation page. The crop details required by the UI include growth conditions, climate, soil, fertilizer, irrigation, pest management, harvesting and season information.

The Search by Location module allows the user to select an Indian state and enter a location. The application then retrieves agriculture recommendations from the database for the selected location. The Location entity stores state, district and locality. Recommendation records are associated with a location and contain a title, type, description and season. Crop or livestock references can optionally be attached when a recommendation is specific to one of them.

The AI chatbot is available after login. To keep the database design consistent with the authenticated UI, Chat_History is included as a small supporting relation that can store the logged-in user's questions and chatbot responses with a timestamp. The AI generation itself remains an application-level service; the database stores only the conversation record if chat history is persisted.

The current UI does not show marketplace, supplier, product-order, farm-activity, weather-dashboard or farm-inventory screens. Therefore those concepts are not included as core entities in this database design. This keeps the database aligned with the implemented interface and avoids adding entities for functionality that is not currently represented by the UI.

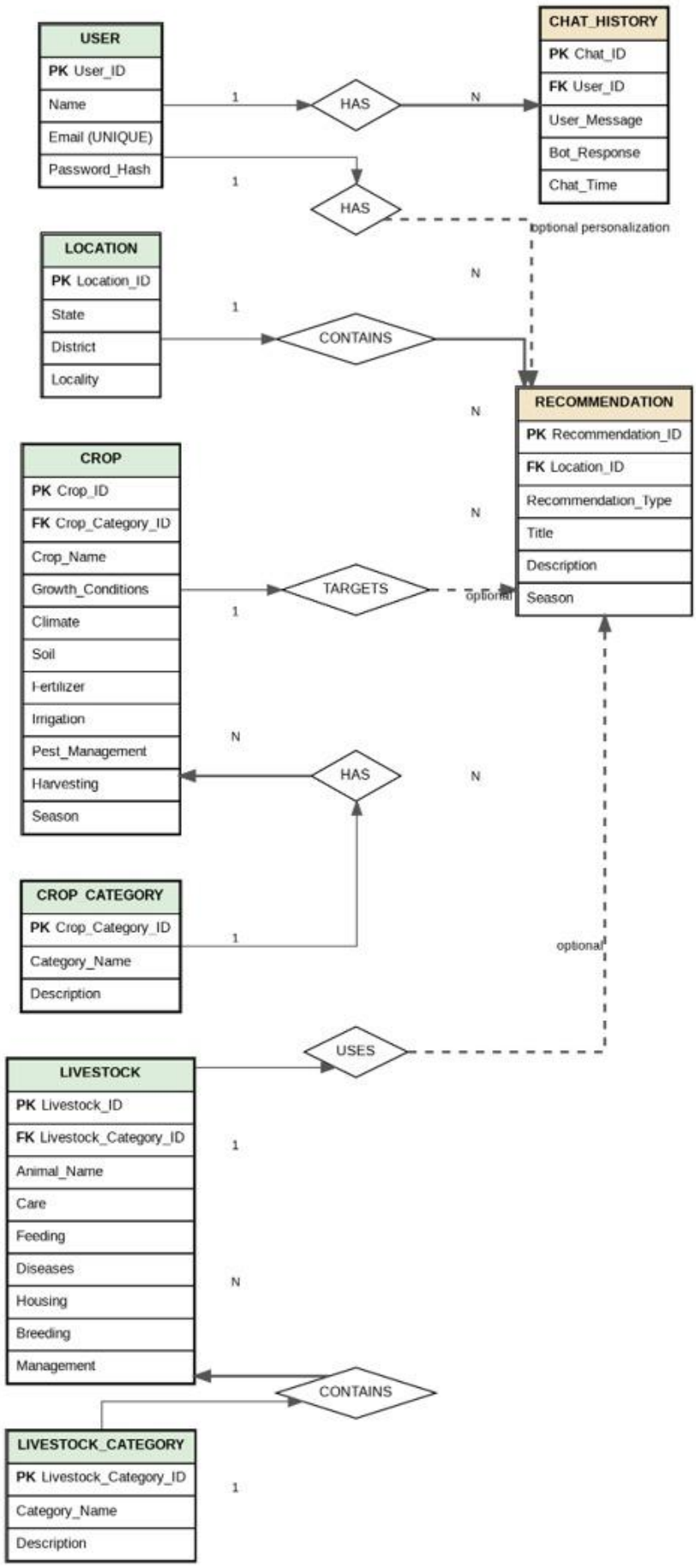
Users

User Type	Responsibilities
Registered User / Farmer	Create account, log in, browse crop and livestock information, search recommendations by location and use the authenticated chatbot.
Administrator / Content Manager	Maintain crop, livestock, category, location and recommendation records in the database.

Main Business Rules

1. Each user has a unique User_ID and a unique email address.
2. A user must have a valid registered account before accessing authenticated pages and the chatbot.
3. Each crop belongs to exactly one crop category; a crop category can contain many crops.
4. Each livestock record belongs to exactly one livestock category; a livestock category can contain many livestock records.
5. A location is identified by Location_ID and stores state, district and locality.
6. A location can have many recommendation records.
7. A recommendation may be general to a location or optionally associated with a particular crop or livestock type.
8. A user can have many chat-history records; each chat record belongs to one user.
9. Crop and livestock detail attributes are stored in their respective entities because the UI displays the complete information on the corresponding detail pages.
10. The state used in Search by Location is restricted to Indian states/locations represented in the database.

3.2. ER Diagram


USER
PK User_ID
Name
Email (UNIQUE)
Password_Hash
HAS
CHAT_HISTORY
PK Chat_ID
FK User_ID
User_Message
Bot_Response
Chat_Time
optional personalization
LOCATION
PK Location_ID
State
District
Locality
CONTAINS
RECOMMENDATION
PK Recommendation_ID
FK Location_ID
Recommendation_Type
Title
Description
Season
CROP
PK Crop_ID
FK Crop_Category_ID
Crop_Name
Growth_Conditions
Climate
Soil
Fertilizer
Irrigation
Pest_Management
Harvesting
Season
TARGETS
optional
HAS
CROP_CATEGORY
PK Crop_Category_ID
Category_Name
Description
optional
USES
LIVESTOCK
PK Livestock_ID
FK Livestock_Category_ID
Animal_Name
Care
Feeding
Diseases
Housing
Breeding
Management
CONTAINS
LIVESTOCK_CATEGORY
PK Livestock_Category_ID
Category_Name
Description
1
N

3.3 Relational Schema

Relation	Attributes
USER	User_ID PK; Name; Email UNIQUE; Password_Hash
CROP_CATEGORY	Crop_Category_ID PK; Category_Name; Description
CROP	Crop_ID PK; Crop_Category_ID FK; Crop_Name; Growth_Conditions; Climate; Soil; Fertilizer; Irrigation; Pest_Management; Harvesting; Season
LIVESTOCK_CATEGORY	Livestock_Category_ID PK; Category_Name; Description
LIVESTOCK	Livestock_ID PK; Livestock_Category_ID FK; Animal_Name; Care; Feeding; Diseases; Housing; Breeding; Management
LOCATION	Location_ID PK; State; District; Locality
RECOMMENDATION	Recommendation_ID PK; Location_ID FK; Crop_ID FK NULL; Livestock_ID FK NULL; User_ID FK NULL; Recommendation_Type; Title; Description; Season
CHAT_HISTORY	Chat_ID PK; User_ID FK; User_Message; Bot_Response; Chat_Time

Relationship-to-Relation Conversion

Relationship	Cardinality	Relational Mapping
User–Chat_History	1:N	User_ID is stored as FK in CHAT_HISTORY.
Crop_Category–Crop	1:N	Crop_Category_ID is stored as FK in CROP.
Livestock_Category–Livestock	1:N	Livestock_Category_ID is stored as FK in LIVESTOCK.
Location–Recommendation	1:N	Location_ID is stored as FK in RECOMMENDATION.
Crop–Recommendation	1:N / optional	Crop_ID is an optional FK in RECOMMENDATION for crop-specific advice.
Livestock–Recommendation	1:N / optional	Livestock_ID is an optional FK in RECOMMENDATION for livestock-specific advice.
User–Recommendation	1:N / optional	User_ID is an optional FK when a recommendation is personalized to a logged-in user.

3.4 Normalization and Functional dependency analysis

3.4.1 Functional Dependencies

Relation	Functional Dependencies
USER	User_ID → Name, Email, Password_Hash; Email → User_ID, Name, Password_Hash.
CROP_CATEGORY	Crop_Category_ID → Category_Name, Description.
CROP	Crop_ID → Crop_Category_ID, Crop_Name, Growth_Conditions, Climate, Soil, Fertilizer, Irrigation, Pest_Management, Harvesting, Season.
LIVESTOCK_CATEGORY	Livestock_Category_ID → Category_Name, Description.
LIVESTOCK	Livestock_ID → Livestock_Category_ID, Animal_Name, Care, Feeding, Diseases, Housing, Breeding, Management.
LOCATION	Location_ID → State, District, Locality.
RECOMMENDATION	Recommendation_ID → Location_ID, Crop_ID, Livestock_ID, User_ID, Recommendation_Type, Title, Description, Season.
CHAT_HISTORY	Chat_ID → User_ID, User_Message, Bot_Response, Chat_Time.

3.4.2 Candidate Keys and Attribute Closure

Relation	Closure	Attributes Obtained	Conclusion
USER	User_ID+	{User_ID, Name, Email, Password_Hash}	User_ID is a candidate key. Email is also a candidate key because it is UNIQUE.
CROP_CATEGORY	Crop_Category_ID+	{Crop_Category_ID, Category_Name, Description}	Crop_Category_ID is a candidate key.
CROP	Crop_ID+	All CROP attributes	Crop_ID is a candidate key.
LIVESTOCK_CATEGORY	Livestock_Category_ID+	{Livestock_Category_ID, Category_Name, Description}	Livestock_Category_ID is a candidate key.
LIVESTOCK	Livestock_ID+	All LIVESTOCK attributes	Livestock_ID is a candidate key.
LOCATION	Location_ID+	{Location_ID, State, District, Locality}	Location_ID is a candidate key.
RECOMMENDATION	Recommendation_ID+	All RECOMMENDATION attributes	Recommendation_ID is a candidate key.
CHAT_HISTORY	Chat_ID+	All CHAT_HISTORY attributes	Chat_ID is a candidate key.

3.4.3 Normalization: 1NF to BCNF

Initial Unnormalized Design

A direct UI-oriented implementation could initially store a crop category and all crop details in one relation:

CROP_INFO(Crop_ID, Category_Name, Category_Description, Crop_Name, Growth_Conditions, Climate, Soil, Fertilizer, Irrigation, Pest_Management, Harvesting, Season)

This design repeats the same category description for every crop belonging to that category, creating update and deletion anomalies.

First Normal Form (1NF)

Each attribute is made atomic and each row represents one crop. Repeating groups are removed. The relation is therefore in 1NF.

Second Normal Form (2NF)

The primary key is Crop_ID. Since the key is a single attribute, partial dependency cannot occur. However, Category_Description depends on the category rather than on the individual crop. Separating category data produces a cleaner dependency structure:

- CROP_CATEGORY(Crop_Category_ID, Category_Name, Description)
- CROP(Crop_ID, Crop_Category_ID, Crop_Name, Growth_Conditions, Climate, Soil, Fertilizer, Irrigation, Pest_Management, Harvesting, Season)

Third Normal Form (3NF)

In the decomposed design, every non-key attribute in CROP depends on Crop_ID, and every non-key attribute in CROP_CATEGORY depends on Crop_Category_ID. There is no transitive dependency among non-key attributes. Hence the relations satisfy 3NF.

Boyce–Codd Normal Form (BCNF)

A relation is in BCNF if every non-trivial determinant is a superkey. In CROP, Crop_ID determines all other attributes and is the candidate key. In CROP_CATEGORY, Crop_Category_ID determines the remaining attributes and is the candidate key. The same pattern applies to LIVESTOCK,

LIVESTOCK_CATEGORY, LOCATION, USER, RECOMMENDATION and CHAT_HISTORY. Therefore the final relations satisfy BCNF under the stated functional dependencies.